Computer Science & IT



C Programming

Structure & Union

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



Topic Structure

Topic Union

Topic

Topic

Topic

Slide

Topics to be Covered



Topic	Dynamic Memory Allocation
Topic	Strategy
Topic	Macro,
Topic	putchar getchar
Topic	

Slide



#define Macro

Parameterized macros:

define Size 100 ← preprocessor

define Max(x,y) x>y? x : y

preprocessor will replace M(x,y) by

x>y? x : y

Slide



#define Macro

Parameterized macros:

Macro's are expanded by preprocessor



#define Macro

```
#include<stdio.h>
#define square(x) x*x
int main(){
      int a, b=3;
      a = square(b+2);
      printf("%d",a);
      return 0;

}
```

Answer: 11

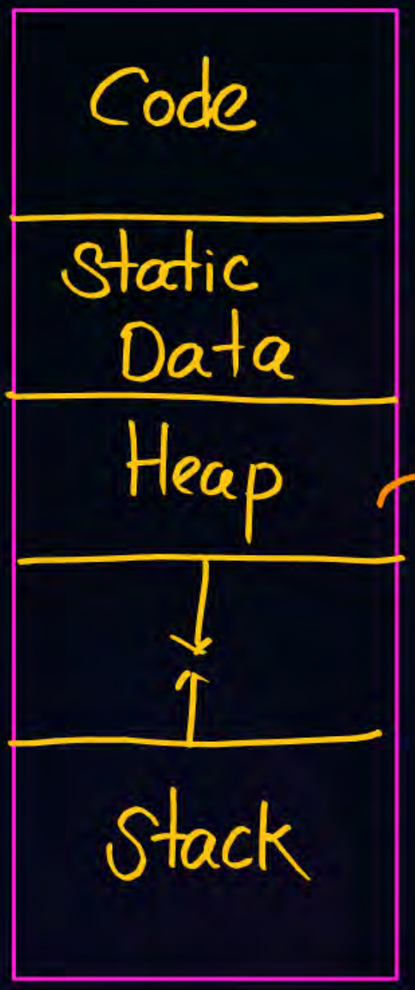
a = b+2 * b+2 ← preprocessor

= 3+2*3+2 → precedence

3+6+2 = 11

Slide

Dynamic Memory Allocation



Runtime :- Dynamic

int a = 10; ← Memory allocation Static at compile time

if you required to allocate memory at run time : Dynamic allocation

User allocate / user deallocate Memory.

Slide



Dynamic Memory Allocation

* Memory allocation done at Runtime called dynamic allocation of memory
* User allocate & deallocate.
* Heap Area of memory is used.



Dynamic Memory Allocation

Memory allocation `#include<stdlib.h>`

1. malloc() (Memory allocation)
2. Single parameter. Size of memory allocated.
3. malloc void pointer return.

Slide

Dynamic Memory Allocation

```
int *pto;
pto = malloc(sizeof(int));
```

pto - 100

sizeof return ammount of memory for data

100



Heap

```
pto = (int*) malloc(sizeof(int));
```

Slide



Dynamic Memory Allocation

* typecast is Not mandetory
  Compiler does automatically for you }

* malloc return void pointer

* No Error if typecasting Not done.

Slide



Dynamic Memory Allocation

ptr = malloc(10 * sizeof(int));

void *ptr;

* void pointer can't be dereference.
* void pointer, pointer anthmetic does not work.

Slide



Dynamic Memory Allocation

calloc() ⟶ Contiguous allocation

calloc(.No. of Locahon, Sizeof(datatype)

char *ch = calloc(75, sizeof(char)); ← 75 Bytes

* Sequential allocation done
* All location inibialized to zero

Slide



Dynamic Memory Allocation

malloc, calloc(), Sizeof(Any data type)

Slide



Dynamic Memory Allocation

malloc()	Calloc()
1 malloc() is have parameter	1 Calloc() there are 2 parameter.
2. void pointer returns	2. void pointer returns
3. Memory is Not initialize	3. All location assigned to zero
4 Any data in sizeof	4 Any datatype allowed sizeof

Slide



Dynamic Memory Allocation

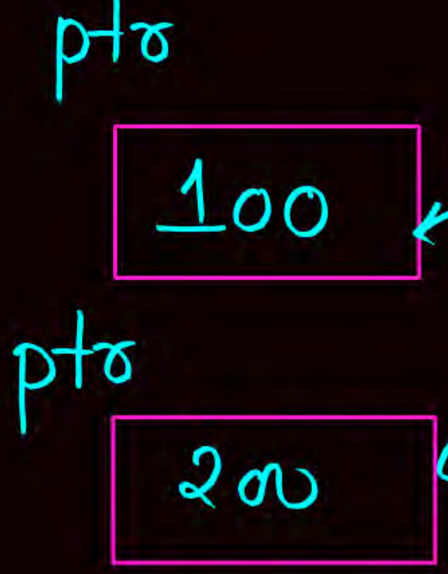
Memory allocation is done by programmer & deallocation is also done by programmer.

function for deallocation free(addres)

```
int *ptr = malloc(sizeof(int));
.....
free(ptr);
```

Slide

if memory allocation done & Not deallocated then its called "Memory Leak"




```
int * ptr = malloc(sizeof(int));
            ↑ address
ptr = malloc(sizeof(int));
free(ptr); ✓
```

Is this leads to memory leak??

yes



Getchar & putchar

Consol : 1 char input . getchar()

getchar :- Even if string is given as input then also first character it takes as input

putchar : outputs a character in terminal.

Slide

Getchar & putchar

2015

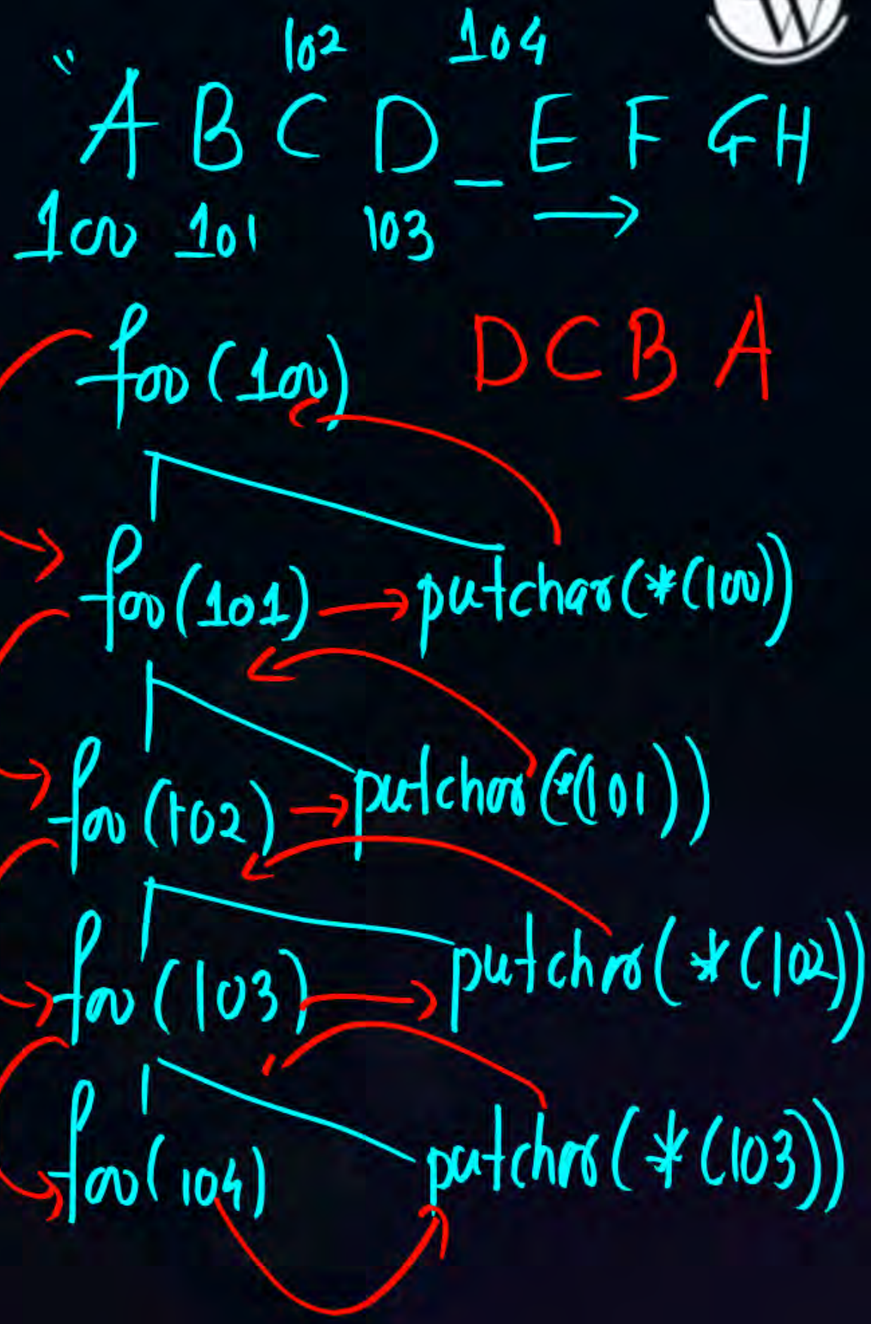
Consider the following function written in the C programming langauge :

```
void foo(char *a)
 {
    if (*a && *a != ' ')
      {
       foo(a+1);
       putchar(*a);
      }
}
```

Not a NULL

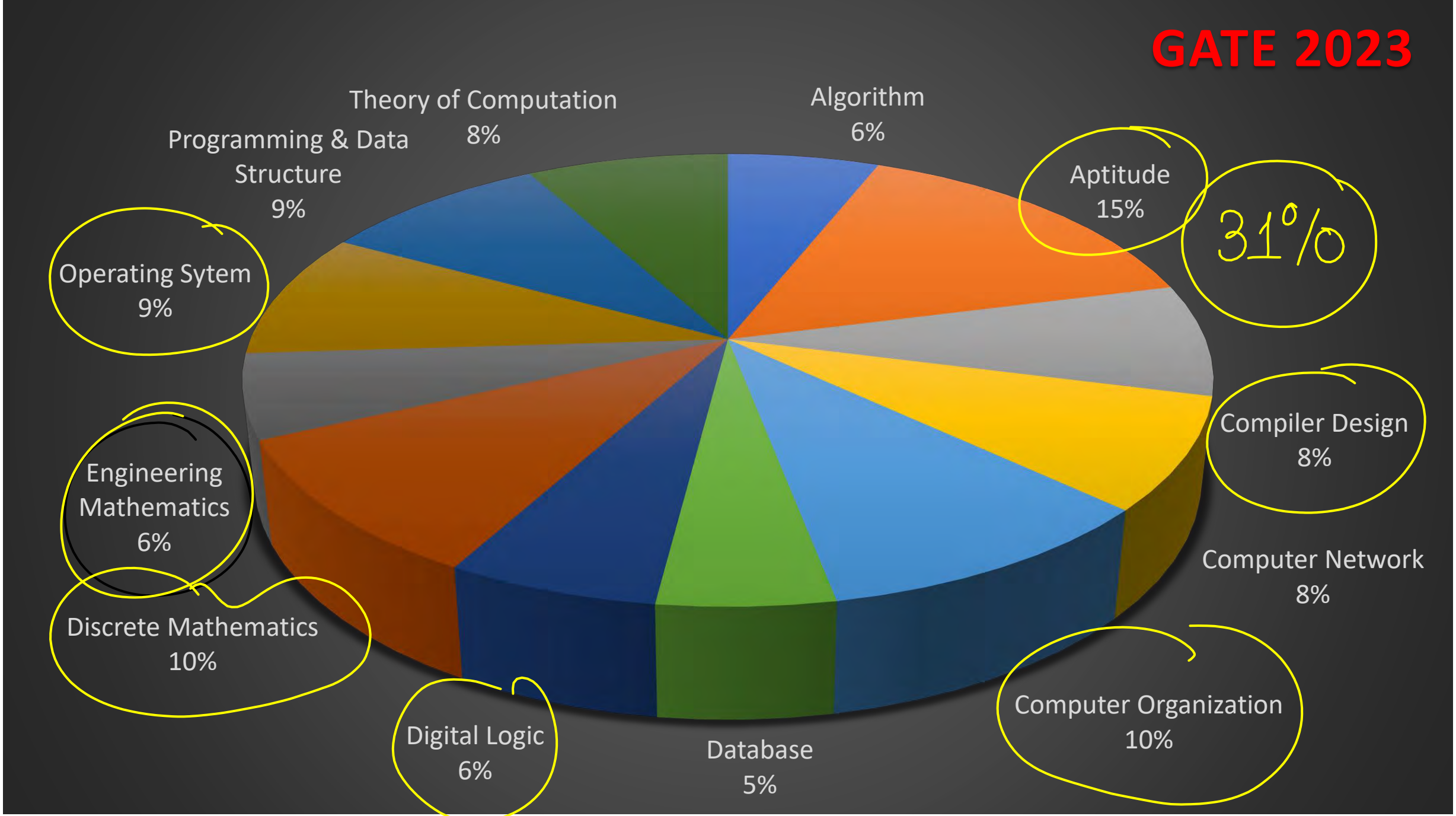
The output of the above function on input "*ABCD EFGH*" is

A. *ABCD EFGH*
B. *ABCD*
C. *HGFE DCBA*
D. *DCBA*

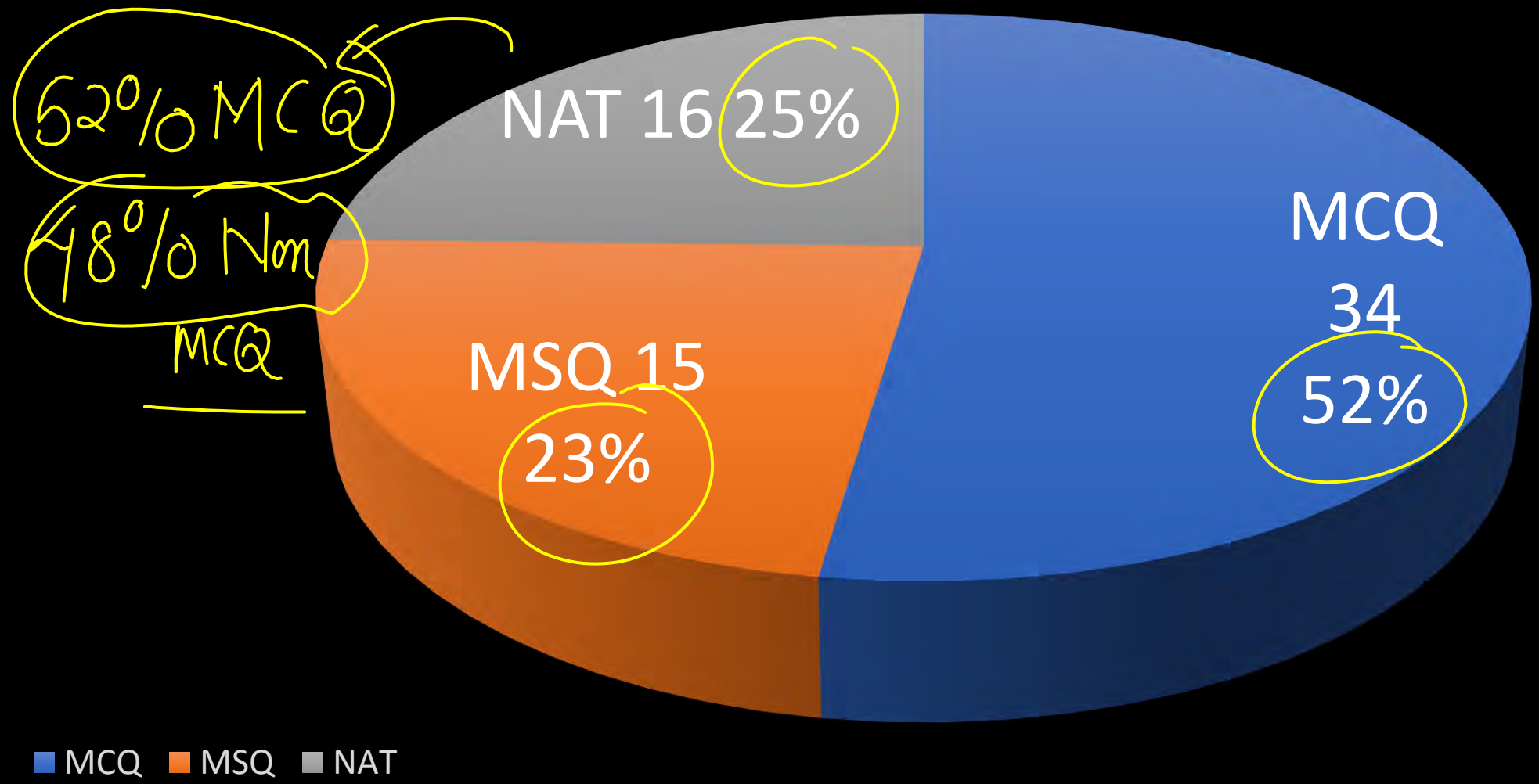


Slide

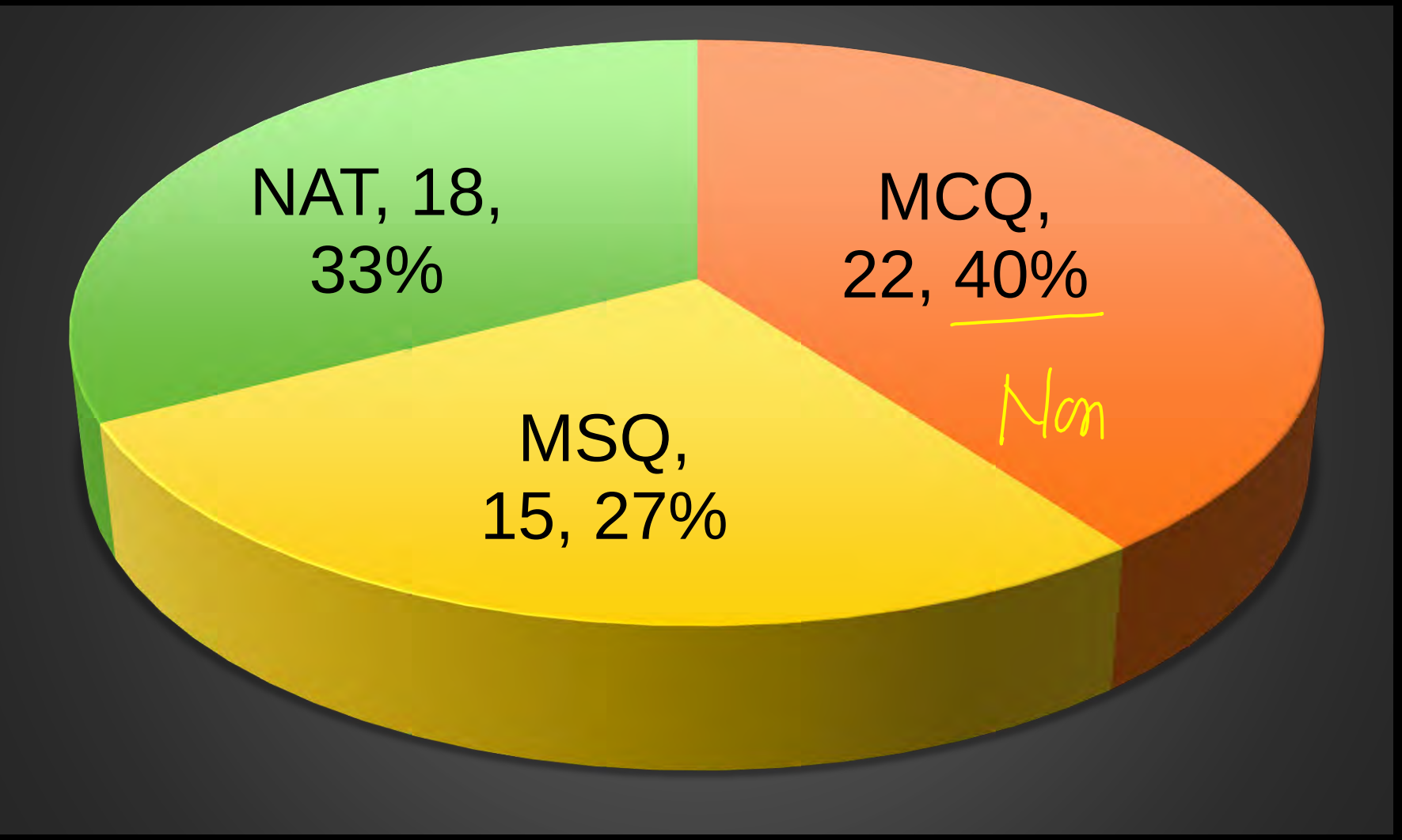
GATE 2023


Theory of Computation
8%
Algorithm
6%
Programming & Data Structure
9%
Aptitude
15%
31%
Operating Sytem
9%
Compiler Design
8%
Engineering Mathematics
6%
Computer Network
8%
Discrete Mathematics
10%
Computer Organization
10%
Digital Logic
6%
Database
5%

GATE 2023 Question Category



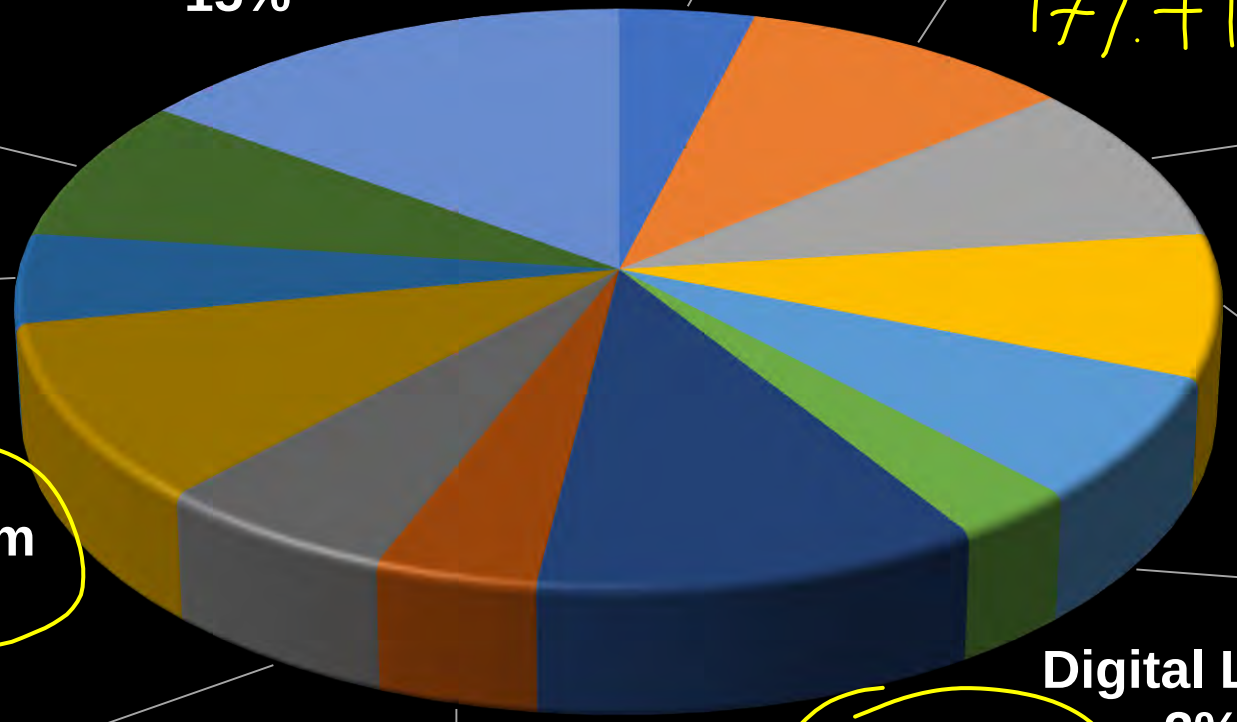
GATE 2022 Question Type Analysis



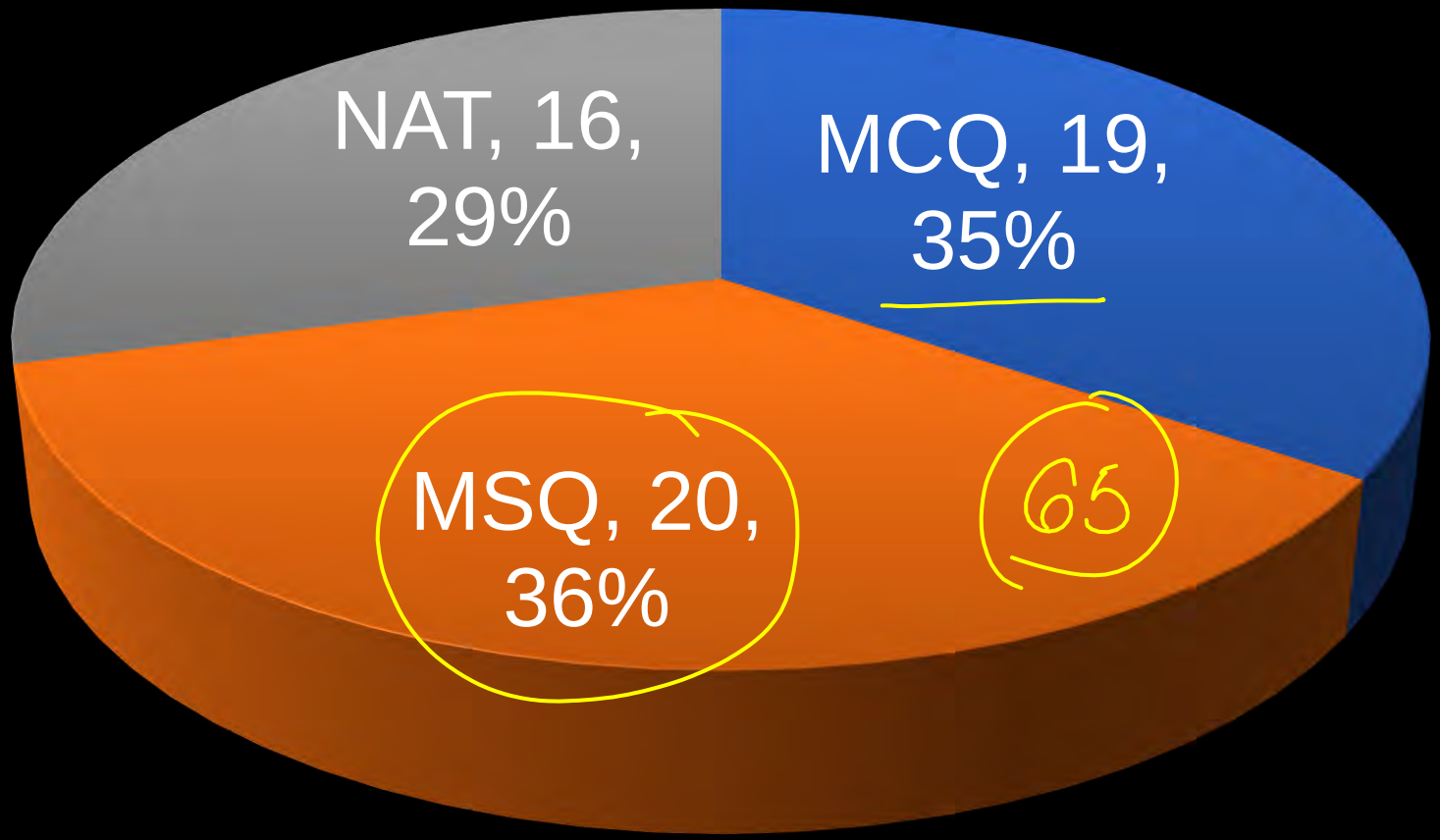
Non MCQ

60%

GATE 2022 Analysis

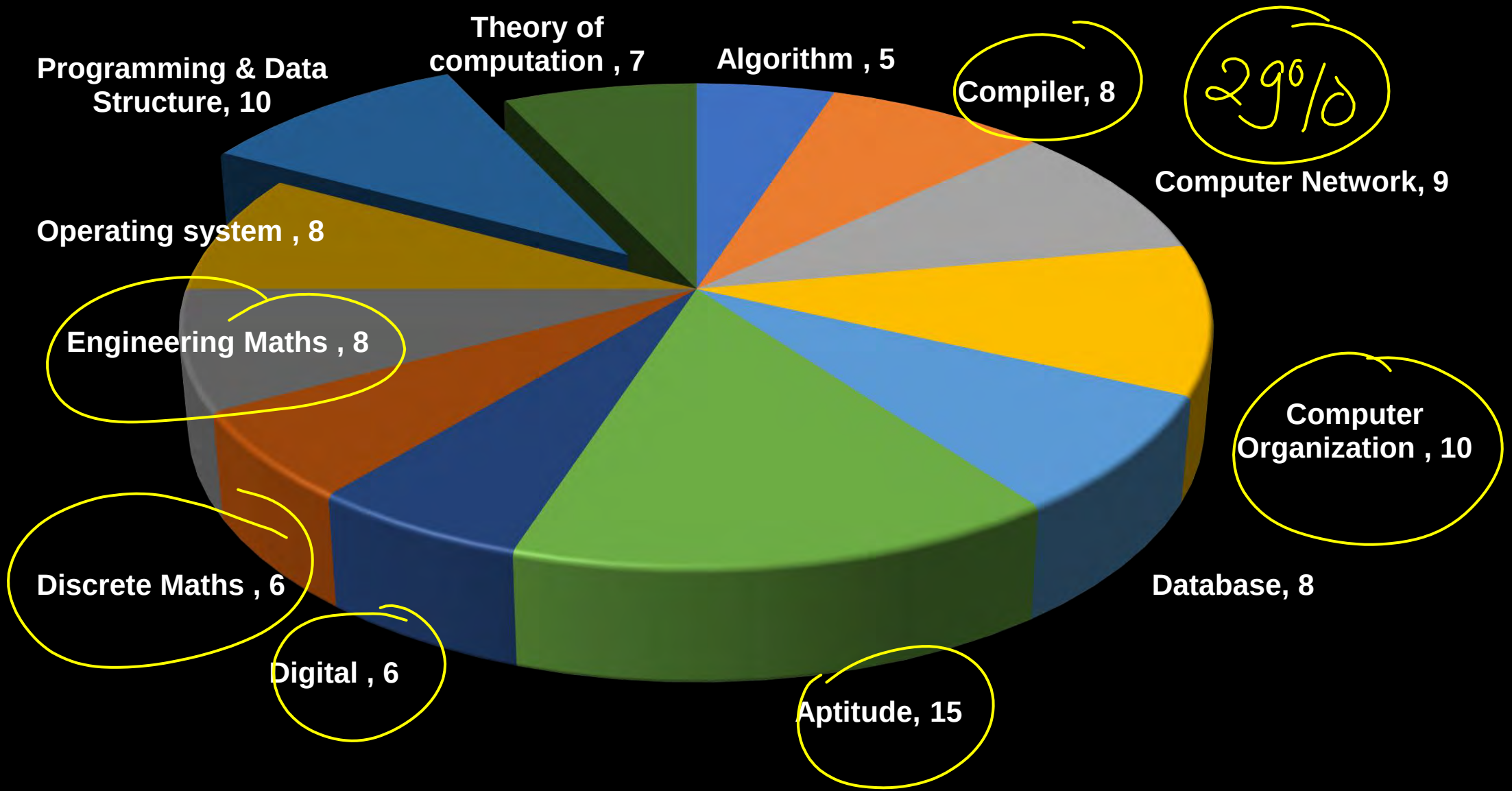


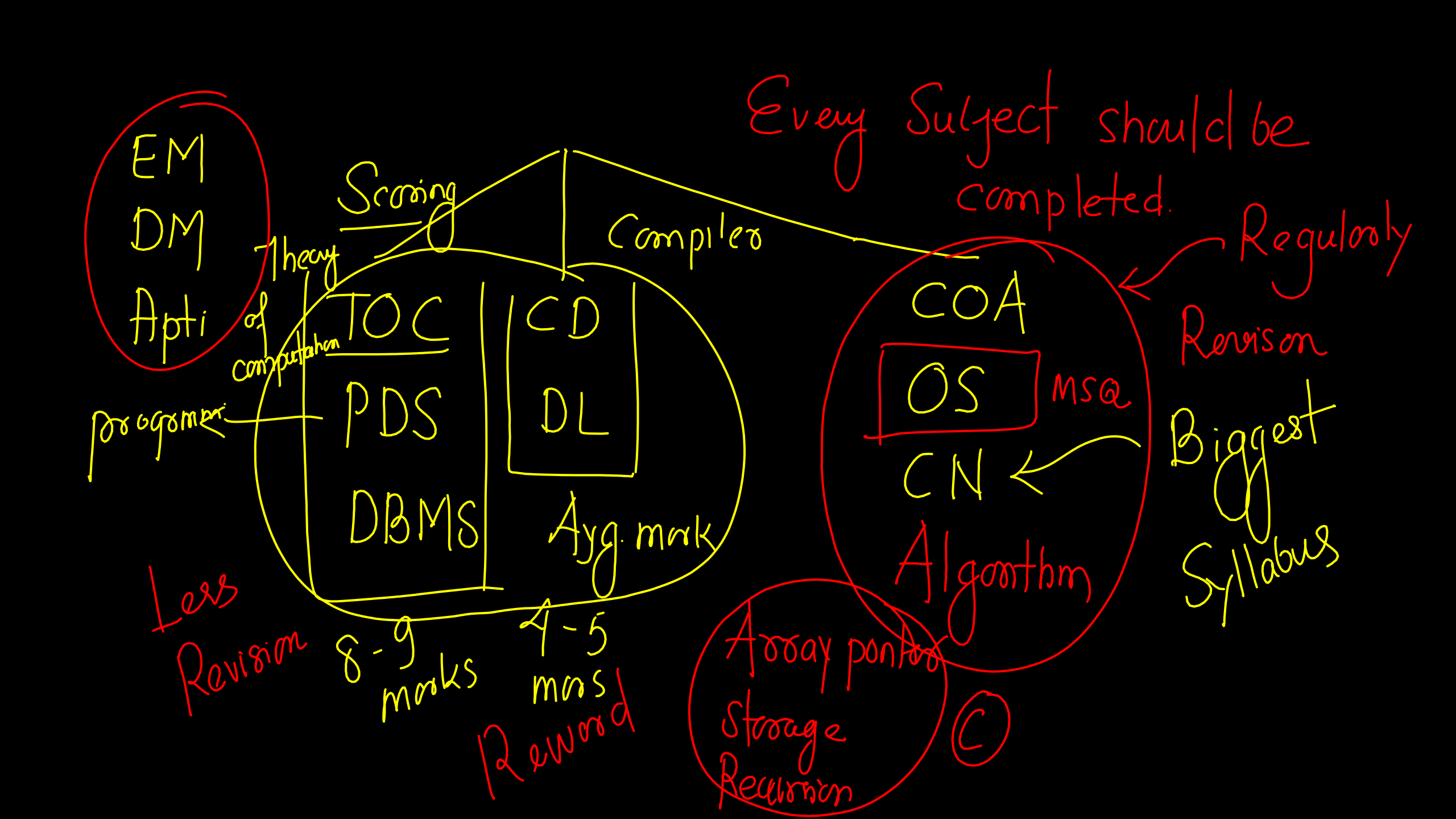
GATE 2024 Analysis



Sat

GATE 2024 Analysis




Every Subject should be completed.
EM
DM
Apti
Scoring
Compiler
Theory of computation
TOC
CD
PDS
DL
DBMS
Ayg. mark
programming
Less Revision
8-9 marks
4-5 mars
Reword
COA
OS
msa
CN
Algorithm
Regularly
Revision
Biggest Syllabus
Array pointer
Storage
Recursion
C

700-800 Hrs

PW

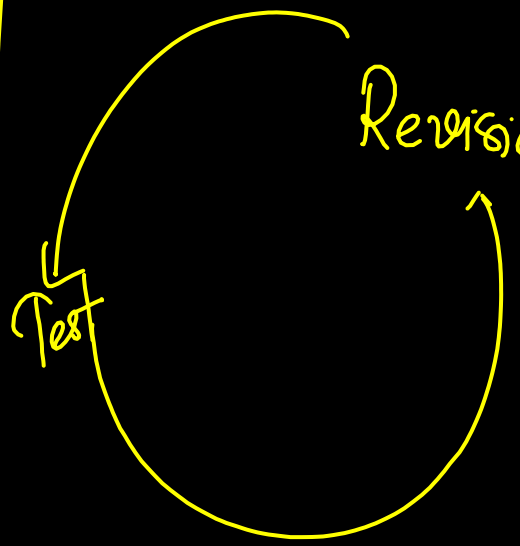
Concept building

November

practice

Test

Revision + Test



At Dec-January



2 mins Summary

Topic — parameterize macro

Topic — Dynamic allocation

Topic — getchar putchar

Topic

Topic

Slide

THANK - YOU

